

Why does stuffing one envelope at a time get the job done faster even though it seems like it would be slower? Because our intuition doesn't take into account the extra time required to sort, stack, and move around the large piles of half-complete envelopes when it's done the other way.² It seems more efficient to repeat the same task over and over, in part because we expect that we will get better at this simple task the more we do it. Unfortunately, in process-oriented work like this, individual performance is not nearly as important as the overall performance of the system.

Even if the amount of time that each process took was exactly the same, the small batch production approach still would be superior, and for even more counterintuitive reasons. For example, imagine that the letters didn't fit in the envelopes. With the large-batch approach, we wouldn't find that out until nearly the end. With small batches, we'd know almost immediately. What if the envelopes are defective and won't seal? In the large-batch approach, we'd have to unstuff all the envelopes, get new ones, and restuff them. In the small-batch approach, we'd find this out immediately and have no rework required.

All these issues are visible in a process as simple as stuffing envelopes, but they are of real and much greater consequence in the work of every company, large or small. The small-batch approach produces a finished product every few seconds, whereas the large-batch approach must deliver all the products at once, at the end. Imagine what this might look like if the time horizon was hours, days, or weeks. What if it turns out that the customers have decided they don't want the product? Which process would allow a company to find this out sooner?

Lean manufacturers discovered the benefits of small batches decades ago. In the post-World War II economy, Japanese carmakers such as Toyota could not compete with huge American factories that used the latest mass production techniques. Following the intuitively efficient way of building, mass production factories built cars by using ever-larger batch sizes. They would spend huge amounts of money buying machines that could produce car parts by